

Emmanuel Ayomide BALOGUN

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EDUCATION

Federal University of Technology, Akure (FUTA) — Ondo, Nigeria

November 2019 – September 2025

B.Tech. Software Engineering, First Class Honours · CGPA 4.54/5.00 (3.792/4.00 *Scholaro equivalent*)

Thesis: Development of a VLM-RAG-Powered Skin Disease Diagnosis and Information System · Supervisor: Olukemi Victoria Olatunde

Relevant coursework: Differential Equations, Linear Algebra, Operations Research, Data Analysis, Data Structures and Algorithms

RESEARCH EXPERIENCE

Undergraduate Research Assistant — Dept. of Software Engineering, FUTA

May 2024 – August 2025

VLM-RAG-Powered Skin Disease Diagnosis and Information System (V-SIDS) · Supervisor: Olukemi Victoria Olatunde

- Fine-tuned LLaVA-1.5 with Low-Rank Adaptation (rank 128, alpha 256) on HAM10000, 10,015 dermatoscopic images across seven pigmented-lesion classes, auto-generating instruction–response training pairs with DeepSeek-VL2, and reached a macro-F1 of 0.89 on a held-out 20% test split against a 0.80 project target.
- Designed a dynamic retrieval layer that replaces a static vector store with live search via the Tavily API, restricted to a trusted-domain allow-list (WHO, Mayo Clinic, WebMD, AAD), injecting the top three passages into the prompt to suppress hallucination.
- Benchmarked the system against GPT-4o and Gemini 2.0 Flash and against ablated variants of itself: factual correctness rose from 0.80 (VLM only) to 0.88 (static RAG) to 0.94 (live web RAG), versus 0.67 for GPT-4o and 0.53 for Gemini 2.0 Flash.
- Ran a 50-case human evaluation on real photographs from volunteers, scored by two independent reviewers: factual correctness 0.94, safety 0.92, clarity 0.86, helpfulness 0.84.
- Deployed the system end to end, fine-tuned model on Hugging Face, FastAPI inference backend, and a Next.js Progressive Web App front end.

CONFERENCES/PUBLICATIONS

Olatunde, O. V., **Balogun, E. A.**, & Omoniyi, V. I. Development of a VLM-RAG-Powered Skin Disease Diagnosis and Information System. *5th African Symposium on Big Data, Analytics and Machine Intelligence (ASBAMI 2026)*, Federal University of Technology, Akure, Nigeria.

WORK EXPERIENCE

Product Manager — [Teksphere Global Services Limited](#), Lagos

January 2026 – Present

- Own the product roadmap for three government-facing platforms delivered to the Joint Revenue Board (JRB) of Nigeria, coordinating seven engineers and designers against regulatory requirements under the JRB Act 2025.
- Led the full lifecycle of the Tax ID System, a dual-facing platform pairing a consumer portal for Tax ID and certificate retrieval (NIN for individuals, CAC number for corporates and cooperatives) with a JRB admin portal for taxpayer management, approvals, certificate issuance, and personnel administration.
- Delivered the E-Receipt System, digitising revenue-collection acknowledgements and cutting paper-based processing time by 95%, and the Fleet Management System, a real-time tracking platform covering 30+ government vehicles, contributing directly to front-end and back-end development on both.
- Translated stakeholder requirements into epics with acceptance criteria across sprints, ran UAT, and served as primary client-facing representative for demos and progress reviews with JRB executives.
- Automated Slack-to-Jira status syncing with an n8n workflow, removing four hours per week of manual status chasing and auto-triggering the testing and UAT process.

Technical Support Engineer — Fuspay Technology Inc., Remote

August 2024 – August 2025

Intern August – December 2024; full-time January – August 2025

- Conducted end-to-end testing of API integrations, performance, and security using Postman across 48+ endpoints, and identified, root-caused, and reported 20+ bugs and performance issues.
- Resolved 78+ critical customer-raised technical issues with engineering and support teams at an average turnaround of three hours, and reviewed 300+ KYC submissions.
- Wrote user guides, API documentation, and product manuals for technical and non-technical audiences, and turned recurring issues into product improvement recommendations.

IT Laboratory Assistant — School of Computing, FUTA

July 2023 – March 2024

- Installed, configured, and maintained 150+ lab workstations and peripheral devices, and provided first-line hardware and software support to 100+ students per semester.

TEACHING EXPERIENCE

Undergraduate Student Tutor — School of Computing, FUTA

January 2022 – June 2024

- Taught first- and second-order differential equations and Laplace transforms to over 40 second-year undergraduates.
- Taught data analysis (statistical sets, probability theorems, linear regression) and operations research (linear programming, the simplex method, transportation and assignment problems) to over 25 students, and trained students in Windows penetration testing for vulnerability assessment.

Physics & Mathematics Teacher, GCSE / SQA National 5 — freelance, remote

July 2022 – May 2025

- Taught Physics and Mathematics to GCSE and SQA National 5 candidates, developing lesson plans and delivering virtual instruction; every student improved by at least one grade band.

Physics & Chemistry Teacher, part-time — Precision College, Akure

February 2020 – April 2020

- Taught Physics and Chemistry to 13 senior secondary students, developed lesson plans, and assessed understanding through assignments, tests, and practical projects.

AWARDS & HONOURS

• Sessional Dean’s List recognition for First Class students, 7 of 10 sessions — FUTA	2020 – 2025
• Visionary Vanguard Award — Association of Computing Students	2024
• Best Class Representative Award — Association of Software Engineering Students	2024

TECHNICAL PROJECTS

Taxman — Tax Filing and Financial Documentation AI Agent

- Built an n8n-based AI agent that files taxes and logs financial documentation from natural language input across four modalities: text, images, voice notes, and documents (PDF/CSV).
- Integrated a RAG system answering tax questions grounded in the Nigeria Tax Act 2025, making every response traceable to current legislation.

Tomato Leaf Disease Detection — Mobile Deep Learning for Smallholder Farmers [code]

- Trained a MobileNetV2 transfer-learning classifier on the PlantVillage tomato subset (~18,000 images, ten classes) with augmentation and dropout, reaching 92% validation accuracy.
- Quantised the model to TensorFlow Lite for offline inference on low-end smartphones, targeting farmers in regions with limited connectivity and no access to agronomists.

LEADERSHIP & VOLUNTEER EXPERIENCE

AR/VR Community Lead — Akure Tech Hub

November 2024 – September 2025

- Organised the hub’s first AR bootcamp using TikTok Effect House, training 20 participants; ran VR training sessions and mentored 87 community members.

President — Association of Computing Students, FUTA Chapter

June 2023 – August 2024

- Led an association of 2,000+ members: partnered with the Google Developer Student Club on an 8-week technical skills bootcamp reaching 250+ students, launched the first AI training academy with about 20 participants, and served as liaison between students and school stakeholders.

Class Representative — Dept. of Software Engineering, FUTA

January 2020 – September 2025

- Represented 62 students as key intermediary with the Head of Department and lecturers, advocating for student interests, shaping departmental policy, and mentoring peers on coursework and exam preparation.

Campus Ambassador — Impact Leaders Club, FUTA

January 2023 – August 2024

- Delivered community outreach programmes aligned with the UN Sustainable Development Goals on poverty eradication and quality education, in partnership with student organisations, NGOs, and government bodies.

Student Legislator — Student Representative Council, FUTA

January 2023 – April 2023

- Authored, debated, and passed two pieces of legislation shaping student union policy, and scrutinised and approved the union’s annual budget.

CONFERENCES & WORKSHOPS

Deep Learning Indaba — August 2025 | DevFest Akure, Google Developer Group — December 2024 | Lisk Africa — April 2024

SKILLS & PROFESSIONAL LICENSES

Technical: Machine learning (algorithm development, fine-tuning, deployment, RAG) | Web development and design | Data analysis and visualisation | API documentation | Cybersecurity (SOC analysis, API security)

Product: Roadmapping | Requirements gathering and prioritisation | Agile/Scrum (Jira) | Stakeholder management | UAT | Client demos and presentations

Languages & Tools: Python, SQL, HTML, CSS, C (beginner) · Git, GitHub, Jira, GitBook, n8n, Postman, Hugging Face Transformers, PyTorch, TensorFlow / TF Lite